

Concept Review

Hardware Interfacing

Why Study Hardware Interfacing?

Having the ability to interface with hardware enables users to safely command a mechatronics device to complete a task. Depending on the mechanical and electrical properties of a system, sending direct voltage commands can have adverse effects. The purpose of this document is to study different techniques which users can adopt when interfacing with hardware to complete a specific task.

Saturation

To ensure safe performance of an actuator different command modification techniques exist. A control system may output a series of commands which could exceed the acceptable range for safe operation. This is where command saturation can be applied. Practically, saturation limits the maximum and minimum values on a signal (see Figure 1). If the signal's value exceeds the maximum, saturation sets the value to the maximum itself, and vice versa for the minimum. In MATLAB/Simulink, a **saturation** block can be used. In Python, the **clip** method from **numpy** can be used.

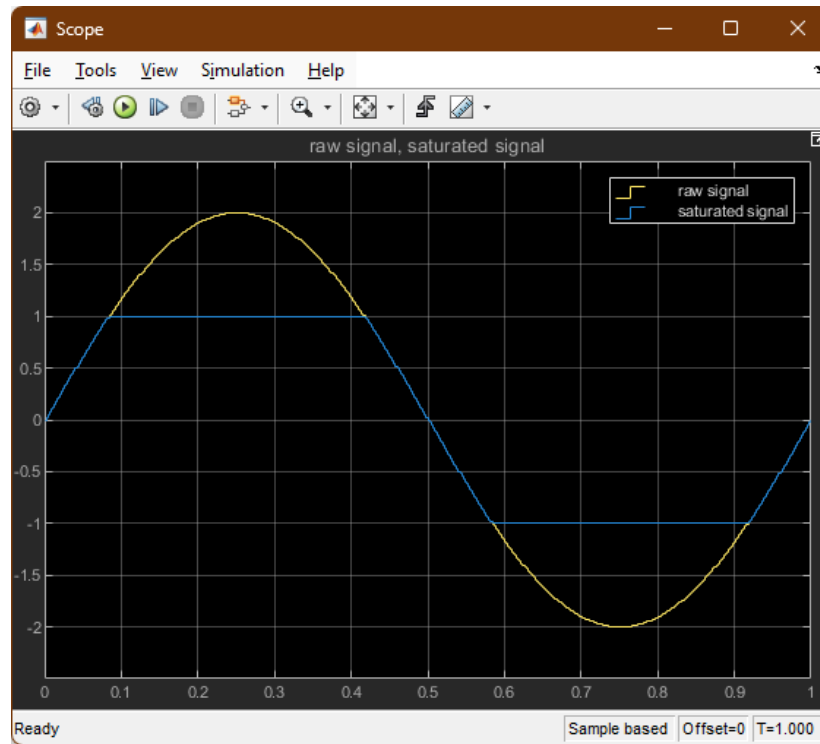


Figure 1: Sinusoid input (yellow) with an amplitude of 2 passes through a saturation set to a maximum amplitude of 1.

Saturation can have numerous advantages depending on the context and where it is applied. In manipulator control, commanding a joint position outside the range of the joint limits may cause damage to the device itself or cause unsafe operation. If a device is continuously applying a command outside of its rated electrical limits, the effect could also propagate to the power supply and cause a brown out, where the overall voltage of the system drops. In this example applying a saturating limit to the position commands w.r.t the maximum and minimum joint limits will prevent this even if the user accidentally commands a value outside this range. Similarly, saturating the PWM command output from a controller with respect to the capabilities of the power supply will ensure that it does not brown out.

Rate-limits

Sending a constant command to a device can visually look like a step input. Sending step inputs can cause large spikes in current which can damage the electronic components of a device. One method for controlling how fast the step input is applied is by rate limitation.

Rate limitation can act like a damper to the input command and control how fast the control input reaches the commanded value (see Figure 2). This can be beneficial in applications involving robotic manipulators, mobile vehicles, smart appliances.

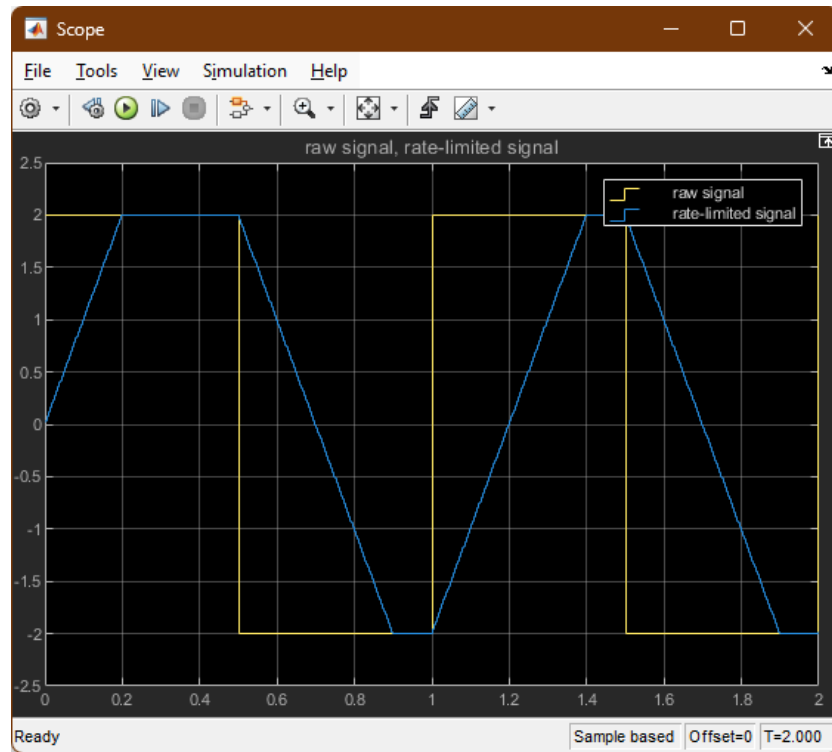


Figure 2. a square wave input (yellow) with an amplitude of 2 passes through a rate limiter set to a maximum rate of 10.

Looking at a robotic arm, to move a joint from -90 to 90 degrees, a controller might see an instantaneous error of 180 degrees. Putting this through a rate limiter of 90 degrees per second allows the desired signal to go through the 180 degrees of change over 2 seconds, improving the controller's position tracking performance. Similarly, passing a discrete change in commanded velocity through a rate limit will saturate the maximum acceleration, providing a gradual speed change. Rate limiting a PWM command also saturates the maximum current draw, thereby limiting the maximum acceleration.

Pulse Width Modulation (PWM)

There are various techniques to change the speed of a motor. An obvious but incorrect method is using a potentiometer to split the motor voltage, which does not work because the motor resistance is much smaller than the potentiometer, and the latter uses up all the power delivered by the voltage source. Pulse Width Modulation (PWM) is a type of signal that consists of switching between high (on) and low (off) at different rates to control the average amplitude of a signal. PWM is a technique used to regulate the effective voltage provided to a motor, essentially turning it on and off fast enough to control the effective voltage, and thus its speed.

The **Duty Cycle** of a pulsing wave is the percentage duration for which the pulse is 'on'. Consider a pulsing wave $V = V(t)$ with Duty Cycle D and time period T , as shown in Figure 3. Using Mean Value Theorem, the average value of this waveform over one period is,

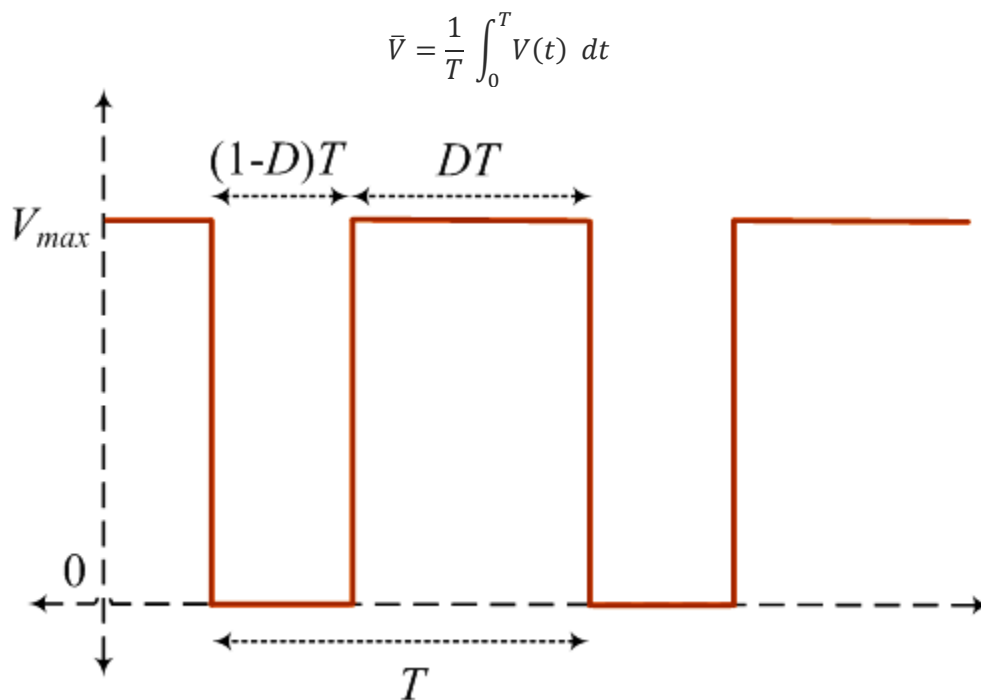


Figure 3: A PWM waveform

If the pulse wave has a maximum value V_{max} for DT s, and minimum value of V_{min} for the remaining time, the average value $\bar{V}$ becomes,

$$\bar{V} = D V_{max} + (1 - D) V_{min}$$

In some applications, the interface to a system requires PWM. For example, servomotors will only respond to a 50 Hz PWM signal, with the duty cycles between 0.5 ms and 2.5 ms representing a minimum to maximum position. Another notable example is **OneShot125** protocols to control a quadcopter's brushless DC motors.

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